Varun Karankar

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Semester 7, Bachelor of Technology (Electronics and Communications Engineering) · **IIIT Allahabad** · Roll No: IEC2023037

Education

•Indian Institute of Information Technology, Allahabad *Expected: 2027*
Electronics and Communications Engineering CGPA: 8.16 (till 6th Semester)

•St. Arnold's School, Indore *2022*
Central Board of Secondary Education, Madhya Pradesh Percentage: 92.4%

Experience

•Selene Care *May - July 2026*
Android Developer Intern

- Architected and deployed a cross-platform Flutter application serving as the core production mobile interface for Selene Care Cosmetics, ensuring seamless web-to-native runtime integration.
- Optimized application performance by implementing advanced WebView resource caching and asynchronous asset loading pipelines, reducing initial interaction latency by 35%.
- Engineered native bridges and secure local persistence layers to handle user session continuity, cart states, and offline fallback scenarios.
- Utilized declarative state management patterns to cleanly decouple business logic from rendering components, enhancing code maintainability.

Projects

•JanShakti.AI *Live: janshakti-ai.tech*
Tech Stack: *Python, TypeScript, FastAPI, VectorDB*

- Engineered a decentralized citizen governance platform leveraging Large Language Models (LLMs) to automate and streamline public grievance redressal workflows.
- Developed an asynchronous FastAPI backend integrating vector embeddings and semantic search pipelines to flag duplicate complaints and route issues dynamically.
- Designed and deployed an intuitive, high-throughput analytics dashboard with transparent public feeds, verified via live production deployment.

•ResilienceAI – Agentic AI Incident Management System
Tech Stack: *Node.js, Express, Python, FastAPI, LangGraph, MongoDB, Redis, BullMQ, Qdrant, React, Socket.IO*

- Designed and developed a full-stack, event-driven incident management system using a microservices architecture with a Node.js/Express REST API and a Python/FastAPI backend connected via BullMQ/Redis.
- Built a deduplication algorithm using regex-based text sanitization and SHA-256 hashing with Redis TTL keys, reducing duplicate alert processing by up to 90%.
- Implemented an AI agent using LangGraph and OpenAI APIs with semantic search (vector embeddings + Qdrant/cosine similarity) to automatically analyze errors against historical data and generate root-cause reports.
- Developed a real-time dashboard with React and Socket.IO, using MongoDB Change Streams to push live updates via WebSockets without polling.

Relevant Coursework

Core Subjects: Object Oriented Programming (OOPS), Database Management System (DBMS), Operating System (OS), Digital Electronics

Positions of Responsibility

- **Coordinator – Tesla Electronics Society:** Spearheaded core technical initiatives, organized robotics/hardware engineering workshops, and managed society event operations.
- **Prayaas – Head:** Mentored over 65+ underprivileged students in fundamental academic subjects.

Achievements

- **Flipkart GRiD 2026:** Secured Semifinalist position in the national flagship engineering challenge.
- **GoBrics Hackathon 2026:** Secured Semifinalist position, ranking in Top 70 teams out of 22,000+ teams across India and Russia.
- **Sankalp Innovation Challenge:** Selected as a National Finalist for JanShakti.AI at the challenge hosted by Ministry of Education, Gov. of India.
- **DSA Instructor – Talentio Academy:** Mentored 3rd & 4th-year B.Tech students in Data Structures & Algorithms; designed and delivered comprehensive lectures on complex data structures, algorithmic optimization, and technical interview preparation.
- **Competitive Programming:** Solved **900**+ problems on LeetCode and Codeforces. Rating: **Knight** (LeetCode).
- **Academic Excellence:** Ranked in top **1%** in JEE Mains 2023 (1.2M candidates) and cleared JEE Advanced.